1.5 Summary

- Some **goals for reading this book** are:
  - adding powerful tools to your coder tool belt, enabling you to write cleaner, faster and more maintainable code
  - improving your ability to spot the opportunity to use known algorithms to speed up your applications
  - learn how to analyze the time and memory impact of an algorithm, and practice this process till it becomes a natural sanity check in your development process
- **Algorithms** should be defined in terms of their input, their output, and a sequence of instructions that will process the input and produce the expected output.
- **Data structures**: a data structure is the concrete implementation of an abstract data type, and that it is made of a structure to hold the data and a set of algorithms that manipulate it.
- **Abstracting a problem** means creating a clear problem statement, and then discussing solution to the problem. This process of moving from a concrete problem to a more abstract problem statement is also fundamental in helping you find the best data structure to efficiently solve the problems you will find in your daily work.
- **Packing your knapsack** efficiently can be tough (especially if you plan to go to Mars!), but with algorithms and the right data structure, (almost) nothing is impossible!